

9th Progress Report

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Outcome of the previous meeting

- Went over the distinction between eligibility and momentum updates, very interesting findings!
- Discussed decision for Flying-V vs Citation in light of IP agreement, awaiting further legal reactions/information on deadline to sign agreement.
- Went over results on IDHP performance, hyperparameter tests. Currently, IDHP takes 30s to finish training on a 60s flight at 50Hz (0.02 s timesteps).
- Talked a little over the Flying-V model, 10-degree elevator deflections to achieve 5 degrees of alpha in the short-period model? Sounds strange but model-matching results seem to suggest that's accurate behaviour.

Description of the results since last meeting

- Got reply from OCW saying no need to apply for a new knowledge embargo form.
- Cleaned up code a little, using a single dictionary to pass all IDHP arguments to the IDHP class, defined functions for plotting and miscellaneous parts of the code and put them in separate files for cleanliness. Somewhat flattened the results folder structure so now it's prettier.
- Did a little rework of the research objective to anticipate possibly moving away from Flying-V research:

Research Objective

To advance the state-of-the-art reinforcement learning based flight controllers, by researching novel sample efficiency improving augmentations to highly-adaptive reinforcement learning algorithms.

- Discovered an issue with eligibility traces. When trying to code the rules for eligibility updates, I realised there was a big problem with incorporating eligibility in neural networks. The gradient of function approximator output wrt it's parameters in the case of neural networks are dependent on existing parameters, e.g. the backdrop operation inevitably means that for weight updates of deep hidden layers, those updates will be functions of weights from shallower layers. Additionally, their formulations have kind-of strict matrix operations where the order of terms highly matter, so trying to compute the back propagation from

output to weight first, and then somehow adding TD error to this backdrop value, becomes tricky. All this leads to the gradient of neural network output wrt hidden layer weights being incompatible with the idea of eligibility, however, there seems to be ways around this as suggested by Kobayashi [1]. In his paper, Kobayashi also recognizes the issue of eligibility traces for neural networks, specifically that the weight updates of hidden layers will depend upon the value of weights from the hallowed layers, which according to him causes the issue that:

... highly parameter-dependent gradients may not be reusable for updating the latest parameters as eligibility traces because they differ significantly from the gradients computed by the latest parameters and violate SGD.

In the case of IDHP, not only does this issue of diverging gradients for deeper layer weights arise, but the output dimensions of the neural networks also pose additional problems. Namely, since the value network approximates value function derivatives wrt system states, the network commonly outputs more than 1 value, i.e. the output layer commonly has 2 or more nodes. This makes computing eligibility trace updates even harder, since the TD error of the value network is thus often a multidimensional vector, so determining the linear algebraic operations to propagate this vector to the deeper layer weight updates becomes even more complicated.

- Added preliminary eligibility augmentation to IDHP, current critic and actor networks only have one hidden layer consisting of 4 neurons, and only the weights between the output layer and the hidden layer use eligibility updates currently, but the performance of augmented IDHP shows marginal improvement over baseline. Average return with eligibility is **-6.39**, while without is **-8.42**, where return is sum of rewards, and reward is defined as difference between reference and actual α for a given timestep. See Figure 1 and Figure 2 for performances of IDHP with and without IDHP.

Plans for next 2 weeks

1. Continue working on eligibility traces, investigate more on how to formulate the matrix operations to propagate TD error to deeper layer weights!!
2. Keep working further on adaptive research questions to remove Flying-V :(
3. Possibly will work on multi-step IDHP, turns out implementation of Ms might be less complicated.

References

- [1] Taisuke Kobayashi. “Adaptive and multiple time-scale eligibility traces for online deep reinforcement learning”. In: *Robotics and Autonomous Systems* 151 (2022), p. 104019. ISSN: 0921-8890. DOI: <https://doi.org/10.1016/j.robot.2021.104019>.

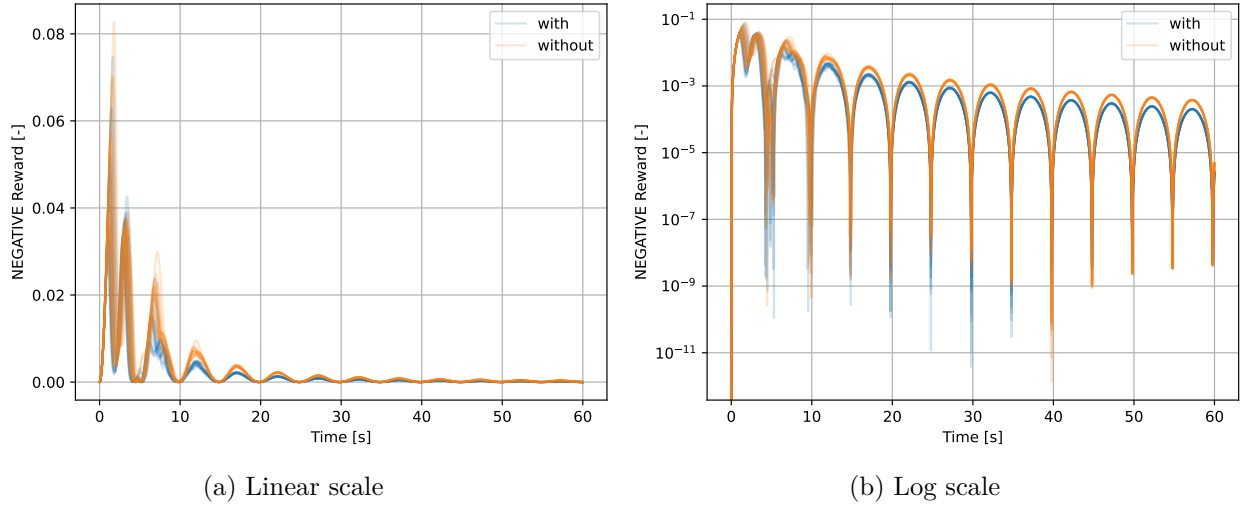
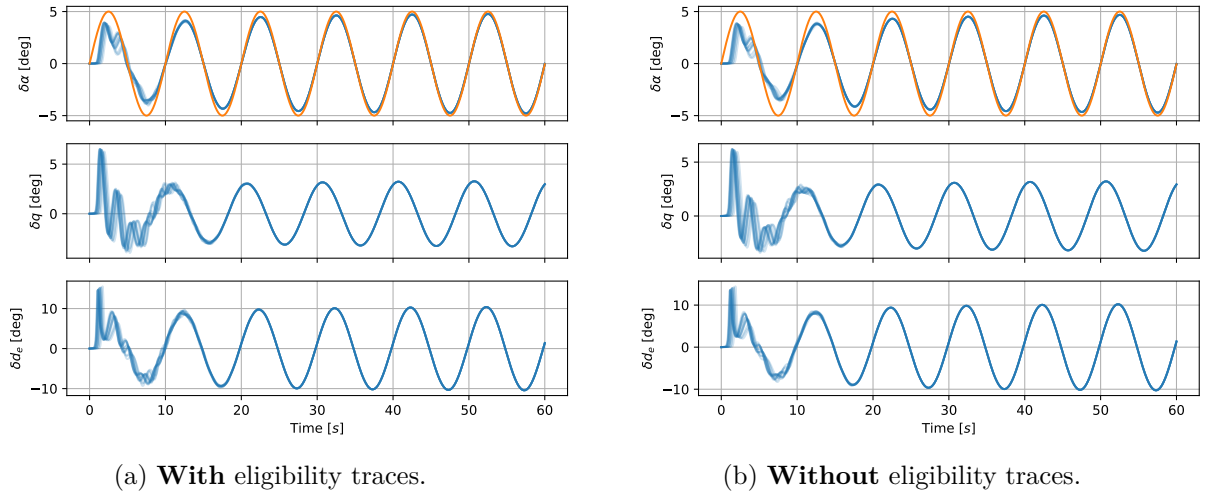


Figure 1: IDHP reward time trace with and without eligibility traces

Figure 2: q, α, δ_e time traces of aircraft under IDHP with and without eligibility traces, orange is reference α , eyenorm of difference in state traces are near-trivial.